

Tutorial 2

Matrices

Q1 Find Matrices A and B

if $A+B = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 2 & 2 \\ 1 & 1 & 2 \end{bmatrix}$ and

$$A-B = \begin{bmatrix} 1 & 4 & 4 \\ 4 & 2 & 0 \\ -1 & -1 & 2 \end{bmatrix}$$

Q2 Find 'X' such that

$$2 \left\{ X + \begin{bmatrix} 2 & -1 & 3 \\ 4 & 2 & 0 \end{bmatrix} \right\} = \begin{bmatrix} -1 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

Q3 Find x, y, z if

$$\begin{bmatrix} 2+x & -1 & 3 \\ 0 & y & z \\ 4 & 1 & 3 \end{bmatrix} + \begin{bmatrix} 1+x & 2 & 3 \\ 0 & 1+y & 4 \\ 2 & 3 & 5 \end{bmatrix} = \begin{bmatrix} 6 & 1 & 6 \\ 0 & -1 & 6 \\ 6 & 4 & 8 \end{bmatrix}$$

Q4 Find value of x & y .

$$\textcircled{1} \quad 2 \left\{ \begin{bmatrix} 3x & -8 \\ 8 & 5 \end{bmatrix} + \begin{bmatrix} 4 & 1 \\ -2 & -y \end{bmatrix} \right\} = \begin{bmatrix} 26 & 0 \\ 12 & 8 \end{bmatrix}$$

$$\textcircled{2} \quad \begin{bmatrix} -3 & x \\ 2y & 0 \end{bmatrix} + \begin{bmatrix} 4 & 6 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 7 \\ -5 & 1 \end{bmatrix}$$

Q5 If $A = \begin{bmatrix} 1 & 3 & 2 \\ -1 & 2 & 0 \\ 4 & 0 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix}$

and $C = \begin{bmatrix} 2 & 1 & 2 \\ 2 & 2 & 1 \\ 1 & 2 & 2 \end{bmatrix}$, then find

matrix D such that $2A - 3B - D = C$

Q6 If $A = \begin{bmatrix} 1 & -2 & 3 \\ -1 & 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 3 \\ 3 & 1 \\ 1 & 2 \end{bmatrix}$

find the matrix $AB - 3I$
where I is the unit matrix of order 2.

Q7 If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ show that
 $A^2 - 4A$ is a scalar matrix.

Q8 If $A = \begin{bmatrix} 2 & 1 & -1 \\ 3 & 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 \\ 2 & 0 \\ 3 & -1 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 3 & 1 \end{bmatrix}$
verify that $(AB)C = A(BC)$.

Q9 If $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 0 & 1 \end{bmatrix}$
 $C = \begin{bmatrix} 2 & -1 & -1 \\ -2 & 2 & 3 \end{bmatrix}$ show that
 $A(B+C) = AB + AC$.

Q15:- Find x, y, z if

$$\left\{ \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & 1 \\ 3 & 1 & 2 \end{bmatrix} + 2 \begin{bmatrix} 3 & 0 & 2 \\ 1 & 4 & 5 \\ 2 & 1 & 0 \end{bmatrix} \right\} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Q10 Find x and y if

$$\begin{bmatrix} 2 & -3 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x & -1 & 2 \\ 1 & 0 & y \end{bmatrix} = \begin{bmatrix} 2x-3 & -2 & 4-3y \\ x+5 & -1 & 2+5y \end{bmatrix}$$

Q11 If $\left\{ 3 \begin{bmatrix} 3 & 1 \\ 4 & 0 \\ -3 & -3 \end{bmatrix} - 2 \begin{bmatrix} 0 & 2 \\ -2 & 3 \\ -5 & 4 \end{bmatrix} \right\} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

Find x, y, z .

Q12 If $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$

Show that $A^2 = A$.

Q13 If I is unit matrix

of order 3×3 and $A = \begin{bmatrix} 1 & 2 & 6 \\ 7 & 4 & 10 \\ 1 & 3 & 5 \end{bmatrix}$

find $A^2 - 3A + I$.

Q14 If $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -3 & 2 \\ -3 & 3 & -1 \\ 2 & -1 & 0 \end{bmatrix}$

Verify that $AB = I$.

Q8

$$C = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$